Excess Returns, Uncertainty Shocks and Intermediaries Balance Sheet

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May 24, 2024

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Motivation

- Empirical evidence suggest strong counter-cyclicality of excess returns of various assets Literature
- Standard rationalization: pro-cyclical SDF
- Recent empirical and theoretical contributions on financial constraints and time varying volatility
- **New hypothesis:** interaction between uncertainty shocks and financial frictions is key in explaining cyclical excess return.

Proposed Mechanism

- Capital ratios of financial intermediaries are significant factor in pricing excess returns Literature
- Intermediaries overweight the downside risk when volatility of assets returns is higher and equity-capital ratios are lower
- Uncertainty shocks triggers flight so safety :
  - $\downarrow$ investment in risky productive assets
  - $\uparrow$ investment in safe securities
  - $\downarrow$ of risky asset prices and $\uparrow$ of excess returns.

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Strategy

- Use 3 different measures of uncertainty:
  - VXO, with shocks identified as in Bloom (2009) Details
  - Financial uncertainty index and macroeconomic uncertainty index (Jurado et al. (2015)), structural shocks identified in Ludvigson et al. (2021) Details
- Specification:

$$y_{t+h} = \alpha_h + \beta^h \epsilon_t + \gamma(L)' \cdot \mathbf{x}_t + u_t$$

$$\mathbf{y}_t = \left[\{R_{it}^{exc}\}_{i \in \{S,B\}}, Credit_t, Treasuries_t, C_t, I_t, Hours_t\right]$$

Local Projections Data Appendix

Excess Returns: Stocks and Bonds

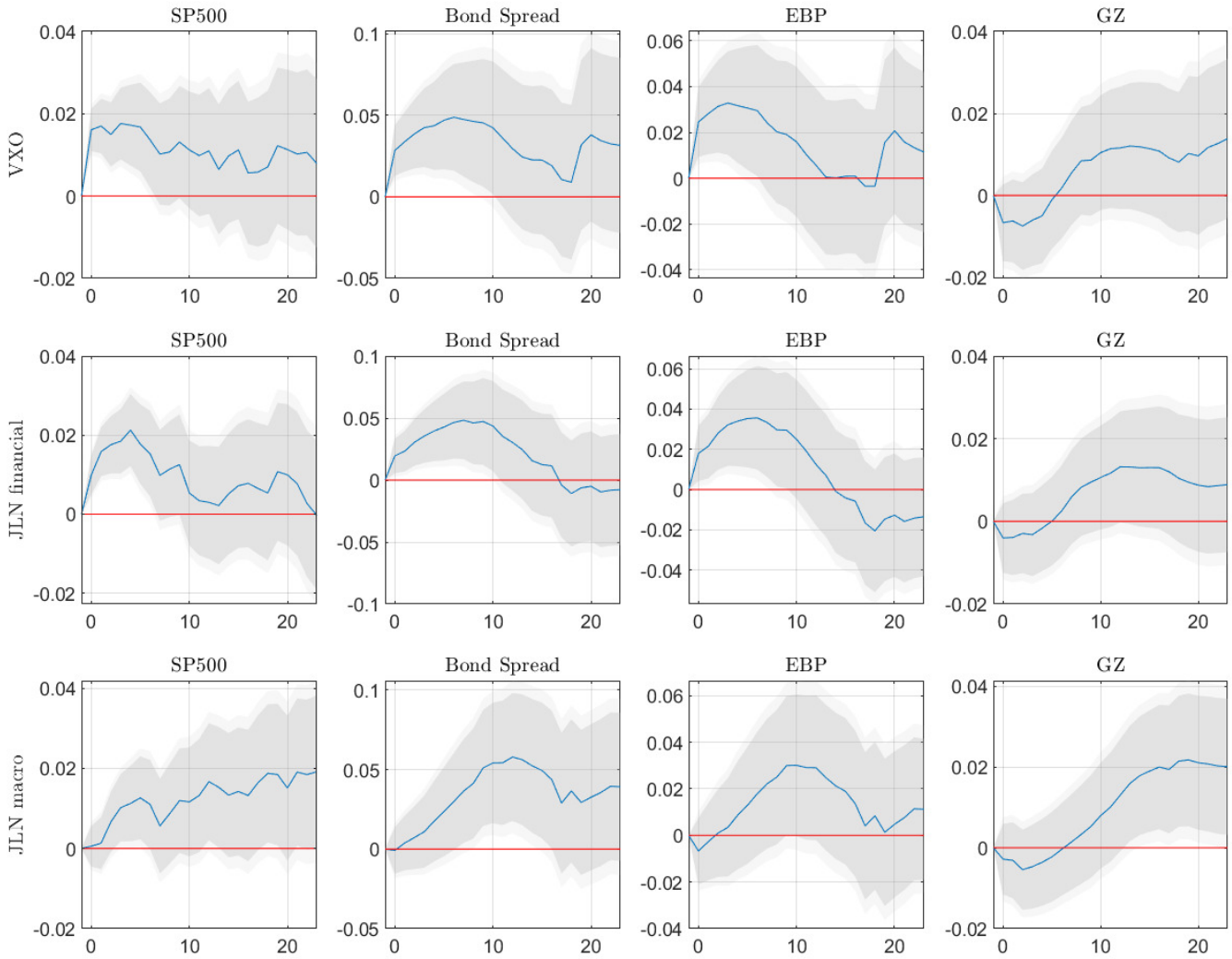

Figure: IRF to 1 s.d. shock. Monthly frequency. Units in p.p.

Financial Sector Balance Sheet

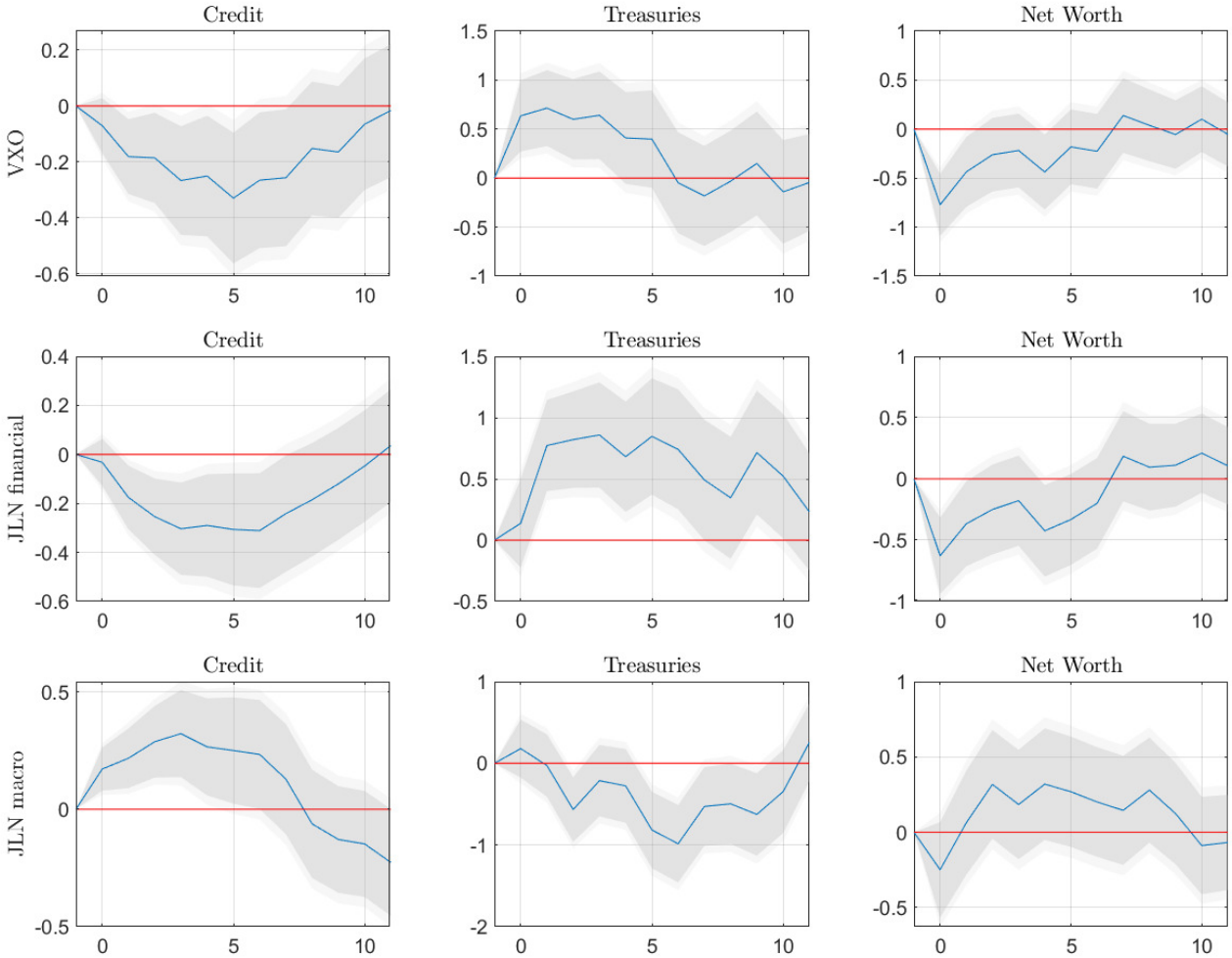

Figure: IRF to 1 s.d. shock. Quarterly frequency. Units in p.p.

Real Variables

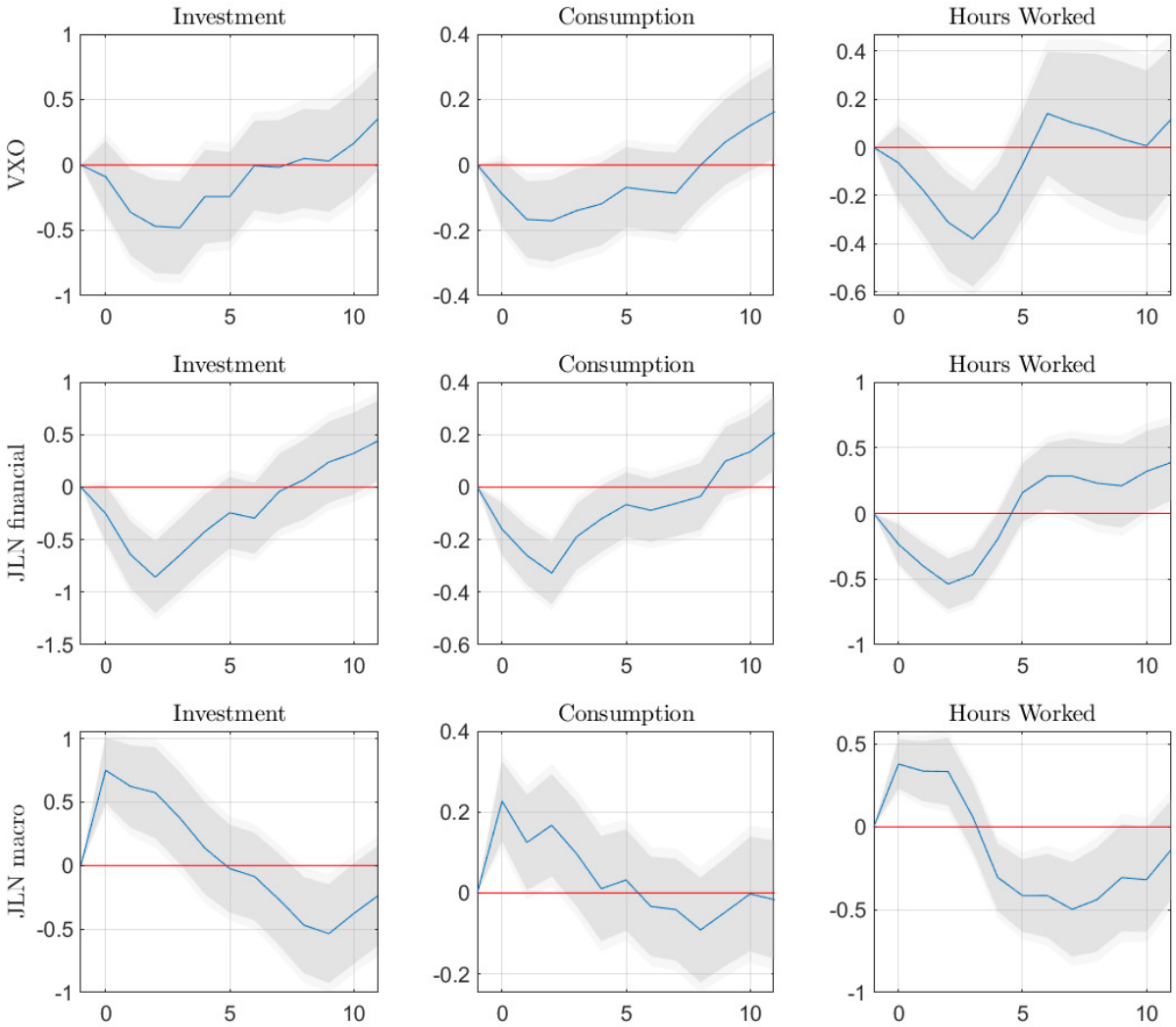

Figure: IRF to 1 s.d. shock. Quarterly frequency. Units in p.p.

Nonlinear Response

- Differential response depending on financial stability of intermediaries in the periods leading to the shock
- Shocks interacted with balance sheet ratios of Primary Dealers (He et al. (2017)) for Stocks, whole financial sector for Bonds Data Appendix
- Specification:

$$y_{t+h} = \alpha_h + \beta_0^h \epsilon_t + \beta_1^h \epsilon_t \mathbb{I}(\Delta n_t < 0) + \boldsymbol{\gamma}(L)' \cdot \mathbf{x}_t + u_t$$

$$n_t = \frac{Equity_t}{InvestedCapital_t}$$

$$IRF(h \mid \Delta n_t < 0) = \beta_0^h + \beta_1^h$$

$$IRF(h \mid \Delta n_t \geq 0) = \beta_0^h$$

Excess Returns: JLN Financial

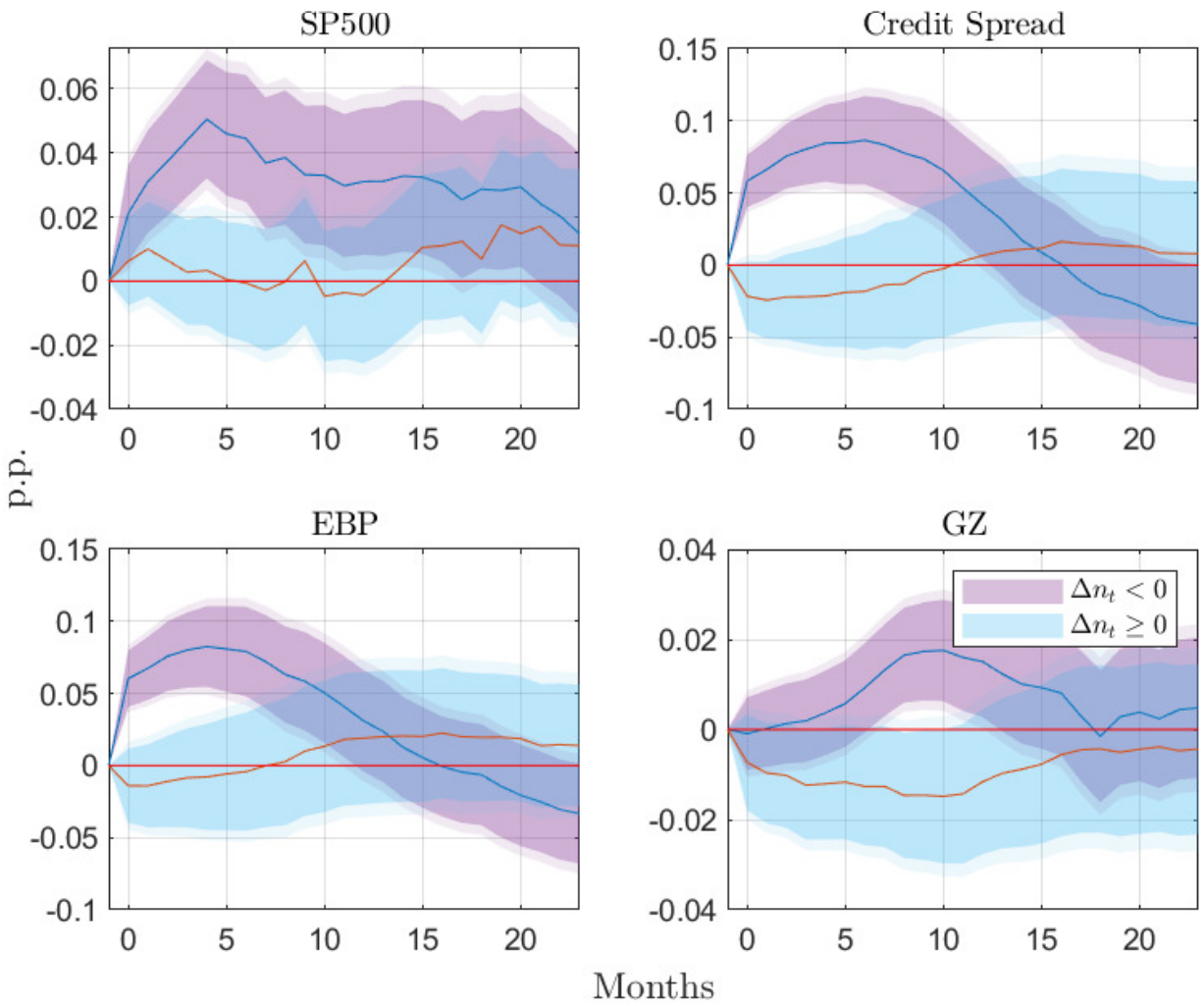

Figure: Purple shows responses to shock following a worsening of capital-equity ratio. Other Measures

Real Variables: JLN Financial

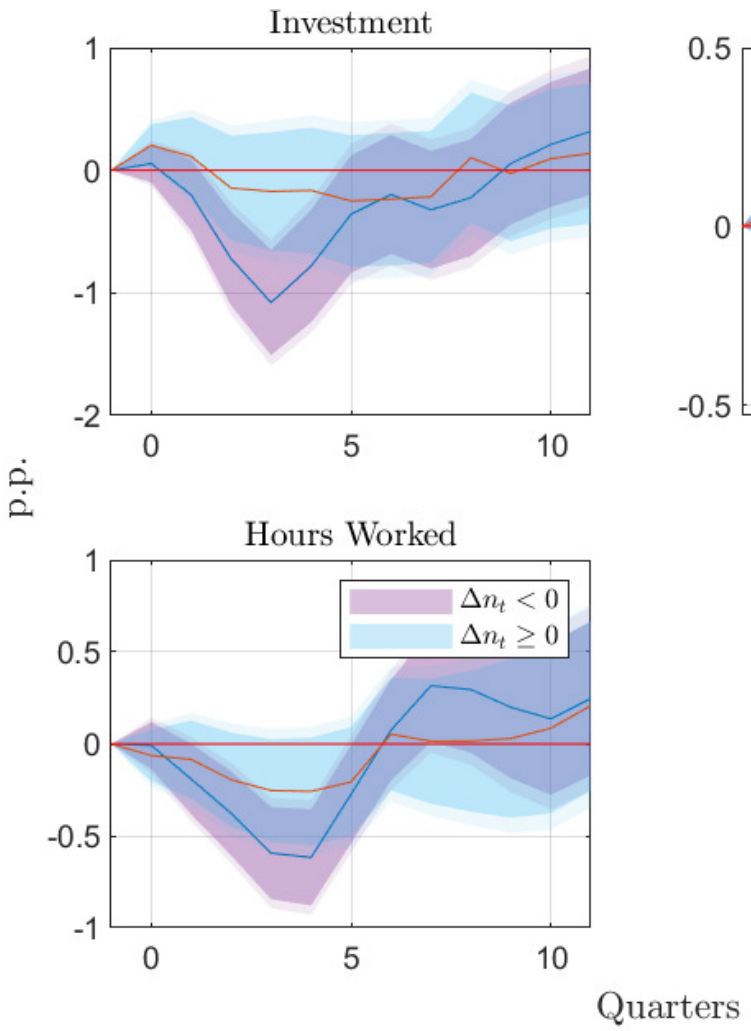

Figure: Purple shows responses to shock following a worsening of capital-equity ratio. Other Measures

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Setup

- Production economy with constrained financial intermediaries (Gertler Karadi 2015);
- HH can hold deposits at a financial intermediary but not K_t directly (gains from intermediation);
- Intermediaries are subject to occasionally binding financial constraints: they can divert a fraction θ of assets for private consumption, hence the ICC:

$$V_t \geq \theta Q_t K_{t+1}$$

- Uncertainty: TFP process

Household

- Preferences:

$$U_t = \max_{C_t, D_t, L_t} \frac{(C_t - \chi L_t)^{1-\gamma}}{1-\gamma} + \beta \mathbb{E}_t U_{t+1}$$

s.t.

$$C_t + D_t = W_t L_t + R_{t-1} D_{t-1}$$

- Euler Equation

$$\mathbb{E}_t[\Lambda_{t,t+1}] R_t = 1$$

- Labor supply

$$W_t = -\frac{\partial U_t / \partial L_t}{\partial U_t / \partial C_t} (= \chi)$$

Intermediaries

$$V_t = \max_{N_{t+1}, K_{t+1}} \mathbb{E}_t\{\Lambda_{t,t+1}[(1-\sigma)N_{t+1} + \sigma V_{t+1}]\}$$

s.t

$$Q_t K_{t+1} = N_t + D_t \qquad \text{Flow of Funds}$$

$$N_{t+1} = R^k_{t+1} Q_t K_{t+1} - R_t D_t \qquad \text{Net worth LOM}$$

$$V_t \geq \theta Q_t K_{t+1} \qquad \text{Financial constraint}$$

Intermediaries

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Optimality Conditions:

$$\mathbb{E}_t \left[\Lambda_{t,t+1}(1 - \sigma + \sigma\psi_{t+1})(R^k_{t+1} - R_t)\right] = \mu_t \theta_t (= 0)$$

$$\psi_t = \frac{\partial V_t}{\partial N_t}$$

Production

- K_t producers

$$Q_t = \Phi'\left(\frac{I_t}{K_t}\right)$$

- Capital LOM

$$K_{t+1} = (1-\delta)K_t + I_t$$

- Final good producer

$$\max_{L_t, K_t} e^{Z_t} K_t^{\alpha} L_t^{1-\alpha} - W_t L_t - Div_t^k K_t$$

s.t.

$$\begin{aligned} W_t L_t &\leq \zeta Q_t K_t \\ Z_t &= (1-\rho_Z)\overline{Z} + \rho_Z Z_{t-1} + e^{v_{t-1}} \varepsilon_t^Z \\ v_t &= \rho_v v_{t-1} + \varepsilon_t^v \end{aligned}$$

Equilibrium

- Goods mkt equilibrium

$$C_t + \Phi\left(\frac{I_t}{K_t}\right) K_t = Y_t$$

- Lab. mkt equilibrium (focus on binding working capital constraint)

$$-\frac{\partial U_t / \partial L_t}{\partial U_t / \partial C_t} = \zeta Q_t K_t$$

- LOM for aggregate net worth:

$$N_{t+1} = \sigma(R^k_{t+1} Q_t K_{t+1} - R_t D_t) + \omega Q_t K_{t+1}$$

Solution Strategy

- Normalize all variables (except prices) by K and denote $x = \frac{X}{K}$
- States: $X = (Z, \log p)$ where $P = RD$
- Eq. Functions $\Theta = (c, R, \psi, Q, i, y, n, p')$
- (c, R, ψ) pin down (Q, i, y, n, p'). Static Equations

Solution Strategy (cont'd)

- (c, R, ψ) come from the solution of the system:

$$\beta c^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(c')^{-\gamma}(1-\sigma+\sigma\psi')((R^k)'-R)\right]=\theta\mu \tag{1}$$

$$\beta c^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(c')^{-\gamma}\right]R=1 \tag{2}$$

$$\psi=\beta c^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(c')^{-\gamma}(1-\sigma+\sigma\psi')\frac{n'}{n}\right] \tag{3}$$

$$\mu(\psi n-\theta Q)\geq 0 \tag{4}$$

Solution Strategy (cont'd)

- Rewrite Equation (3) and condition (4) as

$$\mu = \max\left\{1 - \beta\frac{c^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(c')^{-\gamma}(1-\sigma+\sigma\psi')\right]Rn}{\theta Q}, 0\right\} \tag{5}$$

$$\psi = \max\left\{\beta c^{\gamma}(1-\delta+i)^{-\gamma}\mathbb{E}\left[(c')^{-\gamma}(1-\sigma+\sigma\psi')\right]R, \frac{\theta Q}{n'}\right\} \tag{6}$$

Algorithm

- Start with small value of θ: solution shouldn't be too far from unconstrained (RBC) solution
- Initial guess c_0: RBC solution; R_0 derived from (2) and $\psi_0 = \max\left\{\frac{\theta Q_0}{n_0'}, 1\right\}$
- Given (c_n, R_n, ψ_n) update c by solving for (1) and (5);
- Given (c_{n+1}, R_n, ψ_n), update R using (2) and ψ using (6);

Conclusion

- Strong evidence that the interaction between uncertainty shocks and the financial sector' balance sheet impacts excess returns of various asset classes
- In particular, this interaction can account for excess returns' cyclicality
- Next up: model solution and IRF.

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Literature Review

- **On excess return cyclicality**
  Lustig and Verdelhan (2011), Backus, Routledge and Zin (2008) for stocks, Gilchrist and Zakrajšek 2012 for bonds, Lettau, Maggiori and Weber (2014) for currencies;

- **On uncertainty in macroeconomcs**
  Bloom (2009), Basu and Bundick (2012), Gilchrist et al.(2014), Cesa-Bianchi and Fernandez-Corugedo (2014), Fernández-Villaverde et al. (2011) - (2015) - (2020);

- **On intermediary-based asset pricing**
  Brunnermeier and Sannikov (2014) He and Khrisnamurhy (2012) - (2013), He et al. (2017).
  More from the macro perspective: Bernanke and Gertler (1989), Gertler and Karadi (2011), Gertler and Kyiotaki (2015)

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Bloom (2009)

- Construct volatility index by splicing VXO with volatility of SP500 index
- Run a VAR with

$$\mathbf{y}_t = [R_t, vola_t, ffr_t, cpi_t, wages_t, empl_t, hours_t, ip_t]$$

- Shock identification: ordering and Choleski decomposition
- $vola_t$ shock contemporaneously affects the stock market, then prices and finally quantitites.

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Jurado Ludvigson Ng (2015) and Ludvigson Ma Ng (2021)

- Distinguish between macro and financial uncertainty;
- 132 macro variables divided in 8 groups: income and production, labor mkt, housing, consumption and investment, money and credit, interest and exchange rates, prices;
- 147 financial variables: dividends, prices, d-p ratios, common factors and returns on a large number of portfolios;
- Estimate a FAVAR model for all the variables and compute forecast errors;
- Uncertainty $= \sigma$ of forecast errors
- Shock identification through VAR and sign restrictions.

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Data and Variables

- SP500 returns from CRSP
- GZ spread and EBP from Gilchrist and Zakrajšek (2012)
- Common factors from Kenneth French website
- Credit to nonfinancial businesses by financial institutions (depository institutions and others). Source: BIS statistics. Credit includes both loans and debt securities
- "Domestic Financial Sectors; Treasury Securities; Asset, Level, Millions of Dollars, Quarterly, Not Seasonally Adjusted". Source: FRED
- "Domestic Financial Sector; Total Financial Asset, Level" and "Total Financial Liabilities" to compute net worth. Soruce: FRED
- List of Primary Dealers from NY Fed website; financial ratios from COMPUSTAT.

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Local Projections

The system $(\mathbf{y}_\tau)_\tau$ at time $t+h$ is a (possibly nonlinear) function of the state of the system at time t, $\mathbf{x}_t$, and of the sequence of shocks that hits the system between t and $t+h$:

$$\mathbf{y}_{t+h} = \Phi^h(\varepsilon_{t+h}, \varepsilon_{t+h-1}, \ldots, \varepsilon_t, \mathbf{x}_t)$$

Taking the 1st order Taylor expansion and expectations conditional on the shock vector $\varepsilon_t = \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ v \end{bmatrix}$:

$$\mathbf{y}_{t+h} \approx \Phi^h(\mathbf{0}) + \sum_{j=0}^{h} \frac{\partial \Phi^h(\mathbf{0})}{\partial \varepsilon_{t+h-j}} \varepsilon_{t+h-j} + \frac{\partial \Phi^h(\mathbf{0})}{\partial \mathbf{x}_t} \mathbf{x}_t$$

$$\mathbb{E}_t[\mathbf{y}_{t+h} \mid \varepsilon_t = \mathbf{v}] \approx \Phi^h(\mathbf{0}) + \sum_{j=0}^{h} \frac{\partial \Phi^h(\mathbf{0})}{\partial \varepsilon_{t+h-j}} \mathbb{E}_t[\varepsilon_{t+h-j} \mid \varepsilon_t = \mathbf{v}] + \frac{\partial \Phi^h(\mathbf{0})}{\partial \mathbf{x}_t} \mathbf{x}_t$$

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Local Projections (cont'd)

$$\mathbb{E}_t[\mathbf{y}_{t+h} \mid \varepsilon_t = \mathbf{v}] \approx \Phi^h(\mathbf{0}) + \sum_{j=0}^{h-1} \frac{\partial \Phi^h(\mathbf{0})}{\partial \varepsilon_{t+h-j}} \varepsilon_{t+h-j} + \frac{\partial \Phi^h(\mathbf{0})}{\partial \varepsilon_t} \mathbf{v} + \frac{\partial \Phi^h(\mathbf{0})}{\partial \mathbf{x}_t} \mathbf{x}_t$$

At the same time the expectations conditional on $\varepsilon_t = \mathbf{0}$

$$\begin{aligned}\mathbb{E}_t[\mathbf{y}_{t+h} \mid \varepsilon_t = \mathbf{0}] &\approx \Phi^h(\mathbf{0}) + \sum_{j=0}^{h} \frac{\partial \Phi^h(\mathbf{0})}{\partial \varepsilon_{t+h-j}} \mathbb{E}_t[\varepsilon_{t+h-j} \mid \varepsilon_t = \mathbf{0}] + \frac{\partial \Phi^h(\mathbf{0})}{\partial \mathbf{x}_t} \mathbf{x}_t \\ &\approx \Phi^h(\mathbf{0}) + \sum_{j=0}^{h-1} \frac{\partial \Phi^h(\mathbf{0})}{\partial \varepsilon_{t+h-j}} \varepsilon_{t+h-j} + \frac{\partial \Phi^h(\mathbf{0})}{\partial \mathbf{x}_t} \mathbf{x}_t\end{aligned}$$

Let's define $GIRF(h, t, \mathbf{v}) = \mathbb{E}_t[\mathbf{y}_{t+h} \mid \varepsilon_t = \mathbf{v}] - \mathbb{E}_t[\mathbf{y}_{t+h} \mid \varepsilon_t = \mathbf{0}]$ as in Koop et al. (1986). Then

$$GIRF(h, t, \mathbf{v}) = \frac{\partial \Phi^h(\mathbf{0})}{\partial \varepsilon_t} \mathbf{v}$$

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Static Equations

- $y = e^{Z}(\zeta Q)^{1-\alpha}$
- $i = y - c$
- $Q = \Phi'(i) = a_1 i^{\xi}$
- $n'(1 - \delta + i) = \sigma\{[\alpha y + (1 - \delta)Q] - p\} + \omega Q$
- $p' = R(Q - n')$

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Excess Returns: VXO

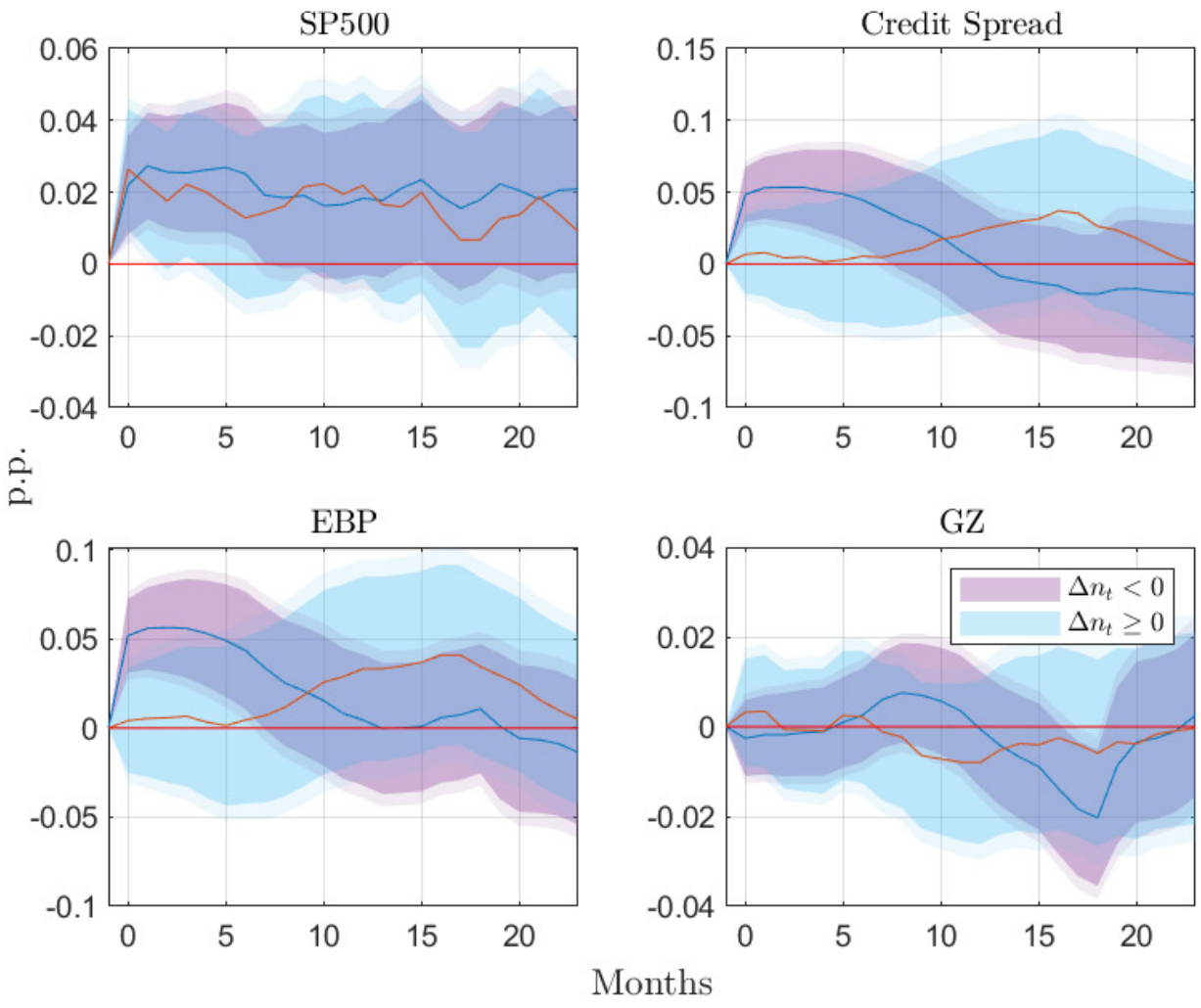


Figure: IRF to 1 standard deviation in financial uncertainty index. Back

Excess Returns: JLN Macro

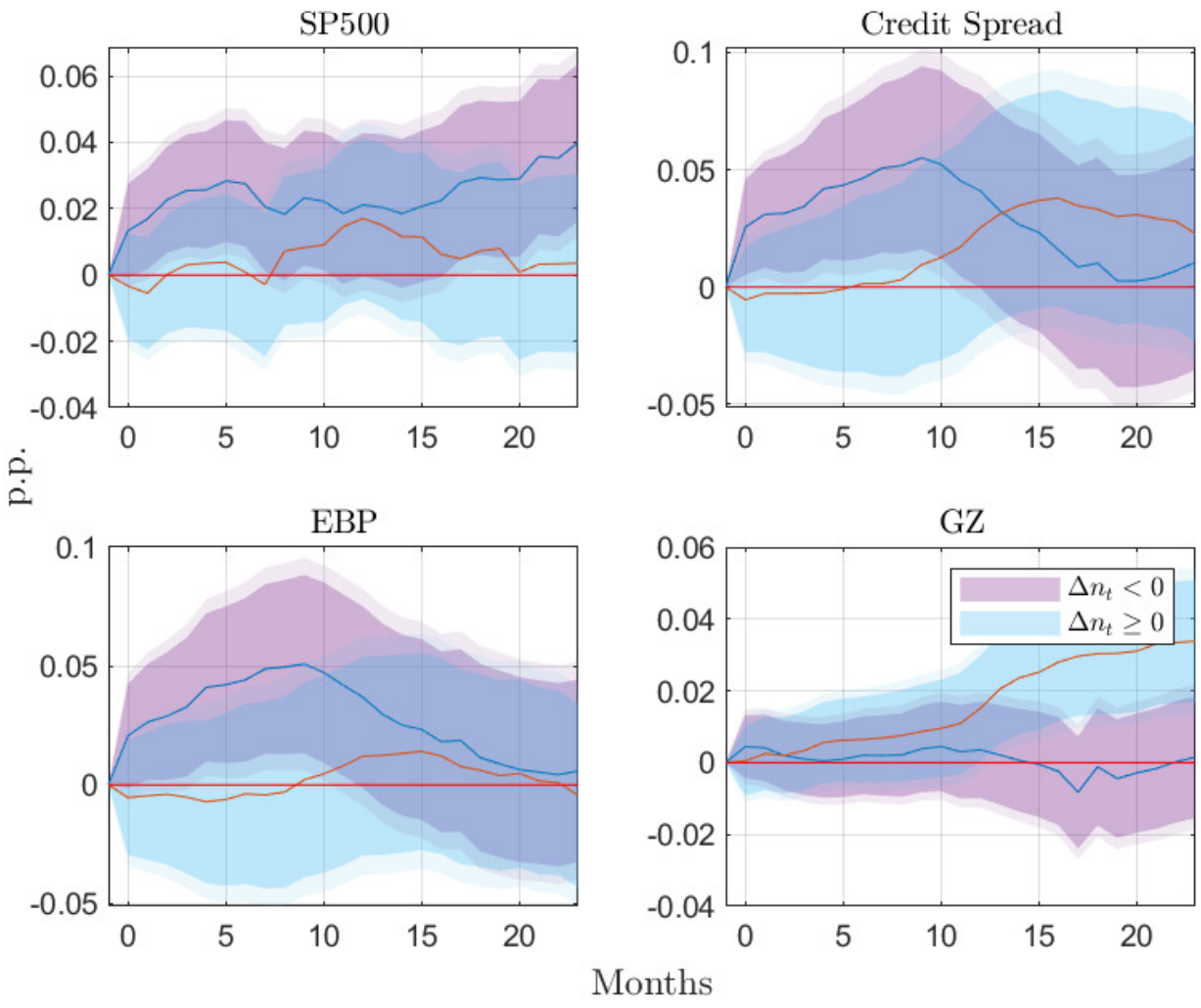
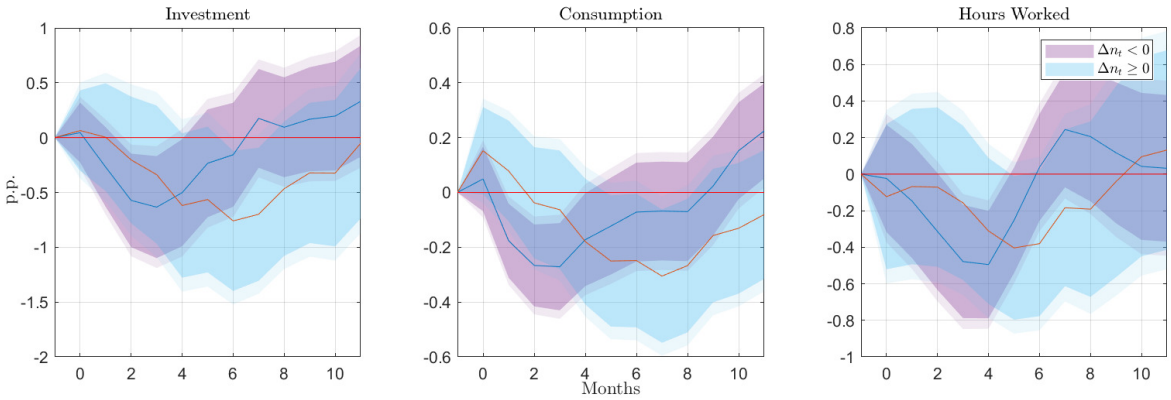


Figure: IRF to 1 standard deviation in financial uncertainty index. Back

Real Variables: VXO



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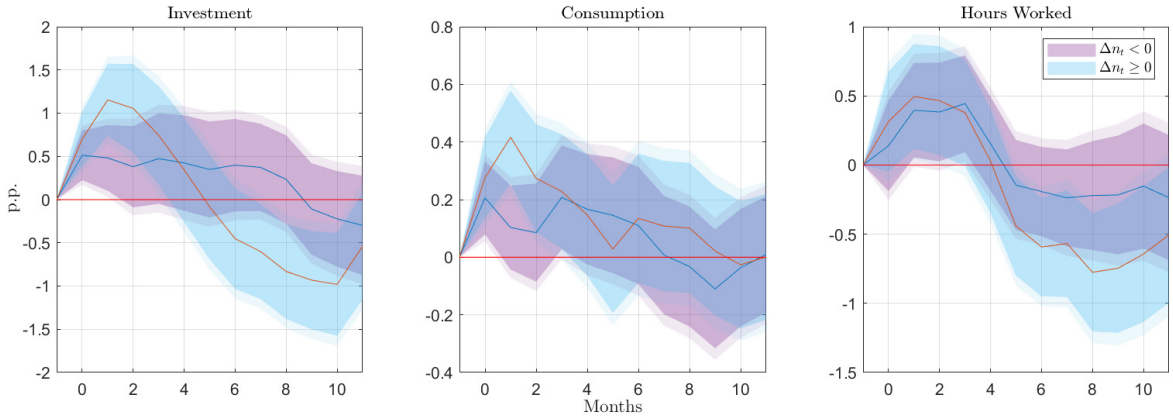

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